

Random Variables, Distributions, and Expectation

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From Outcomes to Numbers

We begin with a probability space:

$$(\Omega, \mathcal{F}, P)$$

- Ω — sample space
- $\mathcal{F}$ — events
- P — probability measure

A random variable is a function:

$$X : \Omega \rightarrow \mathbb{R}$$

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Why Random Variables?

Probability assigns numbers to events.

Random variables assign numbers to outcomes.

They allow us to:

- Quantify uncertainty
- Compute averages
- Define variance and covariance
- Build statistical models
- Define likelihoods in ML

Cumulative Distribution Function (CDF)

For any random variable:

$$F_X(x) = P(X \leq x)$$

Properties:

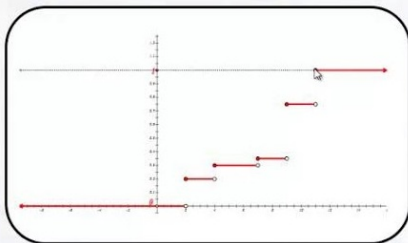
- Non-decreasing
- Right-continuous
- $\lim_{x \rightarrow -\infty} F(x) = 0$
- $\lim_{x \rightarrow \infty} F(x) = 1$

Discrete vs Continuous via the CDF

- Discrete RV $\rightarrow$ CDF has jumps
- Continuous RV $\rightarrow$ CDF is smooth
- Mixed RV $\rightarrow$ combination

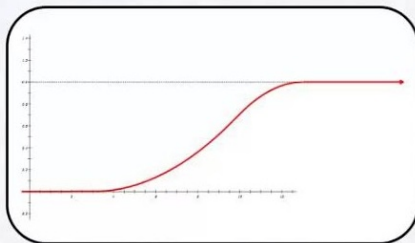
Discrete Distribution

Cumulative Density Function (CDF)



Continuous Distribution

Cumulative Density Function (CDF)



Discrete Random Variables

A random variable is discrete if it takes countably many values.
Probability Mass Function (PMF):

$$p(x) = P(X = x)$$

$$p(x) \geq 0, \quad \sum_x p(x) = 1$$

Relation Between PMF and CDF

$$F(x) = \sum_{t \leq x} p(t)$$

Jumps of the CDF equal the PMF.

Continuous Random Variables

X is continuous if:

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

where $f(x)$ is the probability density function (PDF).

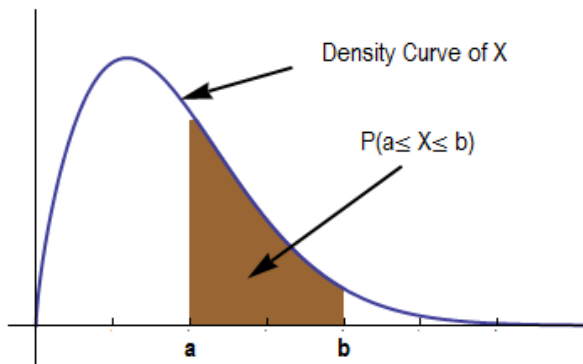
$$f(x) \geq 0, \quad \int_{-\infty}^{\infty} f(x) dx = 1$$

Relation Between PDF and CDF

$$F(x) = \int_{-\infty}^x f(t) dt$$

If differentiable:

$$f(x) = F'(x)$$



Expectation (Discrete)

$$E[X] = \sum_x xp(x)$$

Interpretation:
Weighted average.

Expectation (Continuous)

$$E[X] = \int_{-\infty}^{\infty} xf(x) dx$$

Unified view:

Expectation = integration w.r.t. probability.

Expectation of Functions

For any function g :

Discrete:

$$E[g(X)] = \sum_x g(x)p(x)$$

Continuous:

$$E[g(X)] = \int g(x)f(x) dx$$

$$E[aX + bY] = aE[X] + bE[Y]$$

No independence required.

$$\text{Var}(X) = E[(X - E[X])^2]$$

Equivalent:

$$\text{Var}(X) = E[X^2] - (E[X])^2$$

$$\text{Cov}(X, Y) = E[(X - E[X])(Y - E[Y])]$$

$$\text{Var}(X) = \text{Cov}(X, X)$$

If independent:

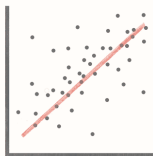
$$E[XY] = E[X]E[Y]$$

Correlation

$$\rho = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}}$$

$$-1 \leq \rho \leq 1$$

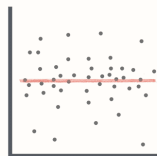
Correlation Coefficient



Positive Correlation



Negative Correlation



No Correlation

Bernoulli Distribution

What it is: Models a single binary trial (success/failure).

$$P(X = 1) = p, \quad P(X = 0) = 1 - p$$

$$E[X] = p, \quad \text{Var}(X) = p(1 - p)$$

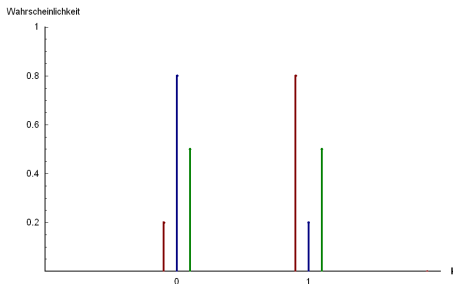


Figure: PMF of Bernoulli distribution

Binomial Distribution

What it is: Models the number of successes in n independent Bernoulli trials.

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

$$E[X] = np, \quad \text{Var}(X) = np(1 - p)$$

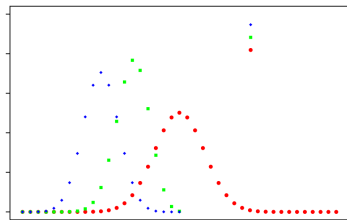


Figure: PMF

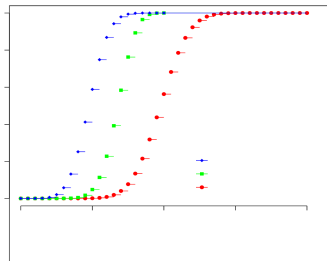


Figure: CDF

Geometric Distribution

What it is: Models the number of trials until the first success.

$$P(X = k) = (1 - p)^{k-1}p$$

$$E[X] = \frac{1}{p}, \quad \text{Var}(X) = \frac{1-p}{p^2}$$

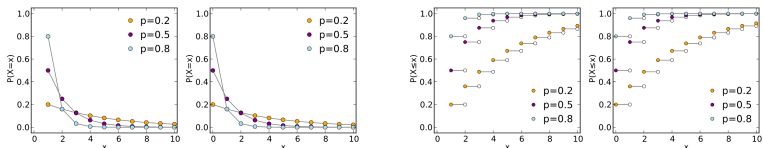


Figure: PMF and CDF of Geometric distribution

Poisson Distribution

What it is: Models counts of rare independent events in a fixed interval.

$$P(X = k) = e^{-\lambda} \frac{\lambda^k}{k!}$$

$$E[X] = \lambda, \quad \text{Var}(X) = \lambda$$

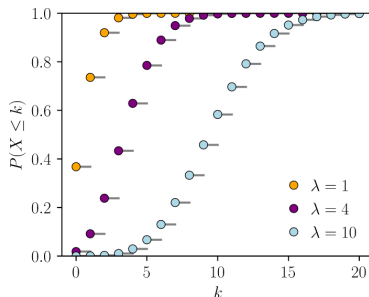
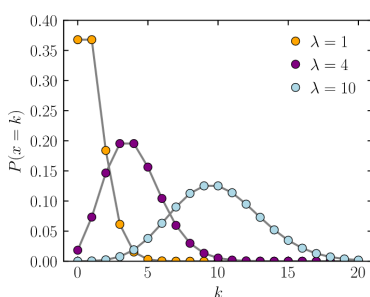


Figure: PMF of Poisson distribution

Uniform Distribution

What it is: Models a variable equally likely anywhere in an interval $[a, b]$.

$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b$$

$$E[X] = \frac{a+b}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}$$

Discrete and Continuous Uniform Distributions: PMF/PDF and CDF

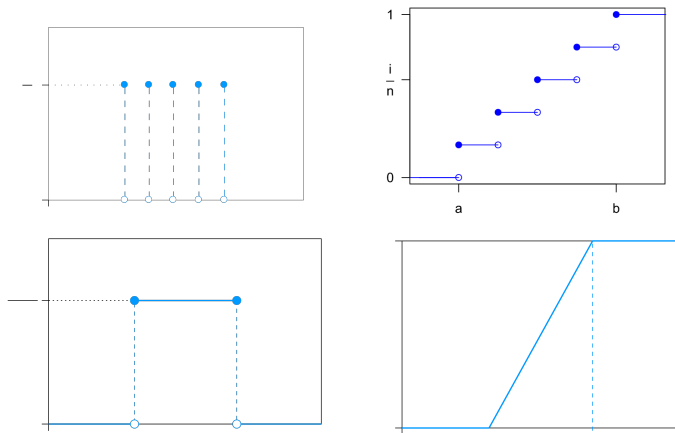


Figure: PMF/PDF and CDF of Discrete and Continuous Uniform distributions

Exponential Distribution

What it is: Models waiting time until the next event in a Poisson process.

$$f(x) = \lambda e^{-\lambda x}, \quad x > 0$$

$$E[X] = \frac{1}{\lambda}, \quad \text{Var}(X) = \frac{1}{\lambda^2}$$

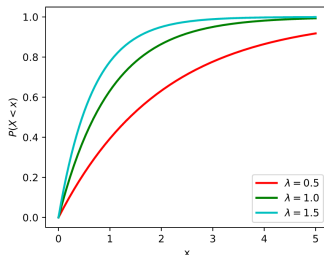
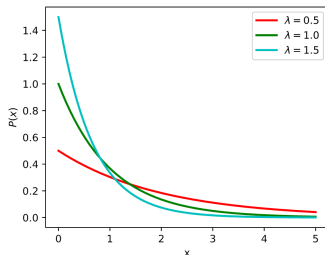


Figure: PDF and CDF of Exponential distribution

Gamma Distribution

What it is: Models waiting time until the k -th event in a Poisson process.

$$f(x) = \frac{\lambda^k x^{k-1} e^{-\lambda x}}{\Gamma(k)}, \quad x > 0$$

$$E[X] = \frac{k}{\lambda}, \quad \text{Var}(X) = \frac{k}{\lambda^2}$$

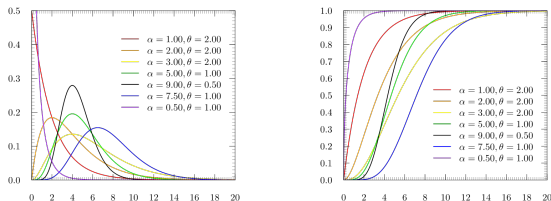


Figure: PDF of Gamma distribution

Beta Distribution

What it is: Models probabilities themselves; defined on $[0, 1]$.

$$f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, \quad 0 < x < 1$$

$$E[X] = \frac{\alpha}{\alpha + \beta}$$

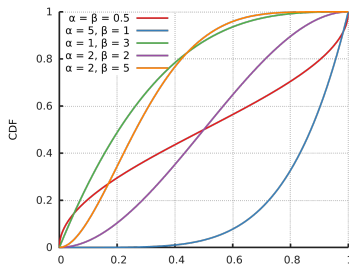
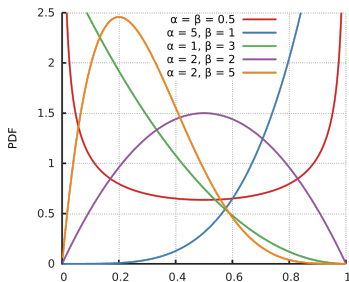


Figure: PDF of Beta distribution

Normal Distribution

What it is: Models symmetric noise and emerges from the Central Limit Theorem.

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

$$E[X] = \mu, \quad \text{Var}(X) = \sigma^2$$

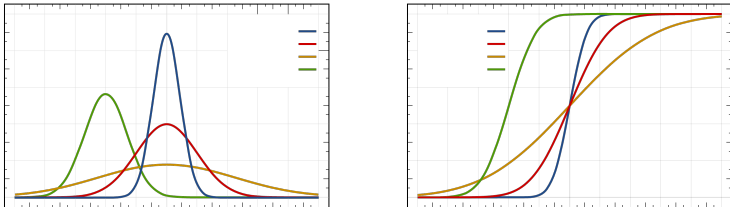


Figure: PDF and CDF of Normal distribution

Joint Distributions

Discrete:

$$P(X = x, Y = y)$$

Continuous:

$$f(x, y)$$

Marginals:

$$f_X(x) = \int f(x, y) dy$$

Covariance Matrix

For vector $X \in \mathbb{R}^n$:

$$\Sigma = E[(X - \mu)(X - \mu)^T]$$

Properties:

- Symmetric
- Positive semi-definite

Likelihood and Loss

Bernoulli likelihood:

$$P(y|x, w) = \sigma(z)^y (1 - \sigma(z))^{1-y}$$

Negative log-likelihood:

$$-\log P(y|x, w)$$

This is logistic loss.

Gaussian Noise $\rightarrow$ MSE

Assume:

$$y = w^T x + \epsilon, \quad \epsilon \sim \mathcal{N}(0, \sigma^2)$$

Likelihood maximization gives:

$$\text{Minimize } \sum (y - w^T x)^2$$

Why Probability is Central

- Loss functions = expectations
- Optimization = minimizing expected risk
- Noise assumptions define models
- Generalization theory built on expectation

Probability is not optional in ML.